

AI Adoption and Productivity: Evidence from UK Firms Using Administrative Data

Cross-Sectional Analysis of the Early Diffusion Phase

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Abstract

Artificial intelligence represents a transformative general-purpose technology with potential to address the UK's persistent productivity challenges, yet empirical evidence on AI's actual productivity impact remains limited, particularly using objective administrative data rather than self-reported measures. Existing studies predominantly focus on US contexts and rely on subjective productivity assessments, creating significant gaps in understanding AI's measurable economic effects within UK institutional frameworks. This study addresses these limitations by providing the first comprehensive analysis linking AI adoption to objective productivity outcomes using UK firm-level administrative data, combining a Department for Science, Innovation and Technology survey of 3,500 businesses with FAME database productivity measures. The analysis employs cross-sectional treatment-control methodology comparing labour productivity between 158 AI adopters and 359 non-adopters across 461 firms, controlling for size, sector, and regional characteristics through progressive specification testing. Results demonstrate productivity premiums of 10-15% for AI adopters that are economically meaningful in magnitude but lack statistical precision, with point estimates suggesting £16,886 higher annual productivity per employee, though confidence intervals encompass zero due to substantial measurement challenges and sample limitations. These findings align with productivity J-curve theory for general-purpose technologies, where early adoption phases involve implementation costs that mask immediate gains, while highlighting fundamental difficulties in measuring digital economy productivity through conventional revenue-based metrics that may inadequately capture AI benefits manifesting through quality improvements and enhanced decision-making processes rather than increased output volumes.

Non-Technical Executive Summary

Executive Summary

This study examines whether firms using artificial intelligence technologies achieve higher productivity than those that do not. The research addresses a critical policy question as the UK government seeks to understand AI's potential contribution to addressing the country's persistent productivity challenges since 2008.

The analysis employs cross-sectional treatment-control methodology combining a survey of 3,500 UK businesses conducted by the Department for Science, Innovation and Technology with administrative financial records from the FAME database. The final dataset comprises 461 firms with complete information, representing diverse sectors and firm sizes, with 158 firms (30.6%) having adopted AI technologies. AI adoption was measured through direct survey questions, while productivity was calculated as annual revenue per employee using objective financial data.

Key Findings

Productivity Differences: AI adopters demonstrated 10.2% higher raw productivity on average before controlling for firm characteristics, equivalent to £16,886 additional revenue per employee. Controlled estimates across multiple specifications suggest productivity premiums ranging from 10.8% to 14.3%. However, these differences were not statistically significant across all models, with confidence intervals encompassing zero, preventing definitive conclusions about AI's genuine productivity impact.

Self-Reported Benefits: Despite inconclusive administrative data, 65.1% of AI adopters reported productivity improvements specifically, suggesting benefits that conventional metrics may not capture effectively.

Adoption Heterogeneity: Strong variation emerges across firm characteristics, with adoption rates of 43.9% among large firms versus 9.5% among micro firms, and sectoral variation from 46.0% in Information and Communication to 16.7% in Transportation and Storage.

Research Limitations and Measurement Challenges

The research highlights fundamental difficulties in measuring AI's economic impact using traditional productivity metrics designed for industrial contexts. Service sector benefits from AI—such as improved decision-making or customer service quality—may not appear in revenue-based measures. Additionally, early-stage AI adoption involves learning costs that can temporarily mask productivity gains, consistent with productivity J-curve theory for general-purpose technologies.

The study faces substantial data limitations. Only 17.0% of survey respondents could be matched with complete productivity data, creating severe sample selection bias toward larger, formal sector firms. Geographic concentration in economically developed regions may not capture AI adoption dynamics in peripheral areas with different infrastructure and market conditions.

Policy Implications

The findings suggest potential productivity benefits from AI adoption while highlighting critical evidence gaps for policy formation. While statistical evidence remains inconclusive, economic magnitudes suggest AI productivity potential warrants continued policy attention. The substantial differences in adoption rates across firm sizes indicate targeted support may be necessary to prevent AI-driven productivity inequality.

The study demonstrates that existing government data collection systems are inadequate for evaluating digital economy policies. Current cross-sectional approaches prove insufficient for credible technology policy evaluation. Future policy development requires enhanced data infrastructure combining administrative records with detailed technology adoption measures, alongside longitudinal studies with improved measurement capabilities.

Conclusion

This UK analysis of AI and productivity provides suggestive evidence of positive effects while revealing substantial methodological challenges. The research establishes baseline understanding of adoption patterns and highlights the need for sustained research investment and methodological innovation as these technologies mature. The path to understanding AI's true productivity impact requires substantial investment in data infrastructure development and longitudinal research commitment to enable robust policy evaluation as measurement systems adapt to digital economy realities.

1 Introduction

1.1 The UK Productivity Puzzle and the Promise of AI

The United Kingdom faces a persistent productivity challenge that has constrained economic growth since the 2008 financial crisis (Haldane, 2018; Castle, Doornik and Hendry, 2020). UK productivity performance has demonstrated consistently weak growth, averaging less than 1% annually over the past decade and creating a substantial divergence from pre-crisis trends (Office for Budget Responsibility, 2017; National Institute of Economic and Social Research, 2024). According to ONS data, from 1970 to 2007 UK average annual multifactor productivity growth was 1.3%, while from 2008 to 2021 it has been just 0.1% (The Productivity Institute, 2024). This productivity stagnation has generated substantial economic losses, with NIESR estimates indicating that had productivity grown at 2% per year since 2008, in line with its historical performance, output per worker would be £5,000 higher today on average (National Institute of Economic and Social Research, 2024). The Office for Budget Responsibility identified that output per hour and output per worker were around 15% below their pre-crisis trend level by 2012, representing a significant constraint on living standards and fiscal sustainability (Office for Budget Responsibility, 2017).

International comparisons highlight the severity of this challenge. UK productivity levels lag substantially behind France, Germany, and the United States across multiple sectors (O'Mahony and Ark, 2003; Office for National Statistics, 2023). In 2021, the UK's output per hour worked was £46.92, compared to £58.88 in the US, £55.83 in Germany and £55.50 in France, representing productivity gaps of 20% below the US and approximately 16% below France and Germany (Office for National Statistics, 2023; Bloomberg, 2023). The UK ranked 31st out of 35 OECD countries in growth of output per hour from 2008 to 2017, with particularly pronounced gaps in knowledge-intensive industries where digital technologies might be expected to generate the greatest returns (Centre for Economic Policy Research, 2020). The Office for Budget Responsibility consistently emphasises that sustained productivity growth remains critical for long-term fiscal sustainability and the achievement of broader economic policy objectives, including the government's growth mission (Parliamentary Office of Science and Technology, 2025).

Artificial Intelligence represents the latest in a sequence of general-purpose technologies that have historically transformed economic structures, following steam power, electricity, and information and communication technologies. Economic theory suggests AI possesses the fundamental characteristics necessary for productivity transformation: pervasiveness across multiple sectors, continuous technological improvement through algorithmic advancement, and innovation complementarities that enhance research and development processes across diverse applications.

Previous technological transformations demonstrate that the relationship between innovation and measured productivity exhibits systematic temporal delays. For example, the "productivity paradox" of the 1980s and 1990s, where substantial ICT investment failed to generate measurable productivity gains until the late 1990s. Solow's observation that "computers are everywhere except in the productivity statistics" documented this phenomenon, which required organisational adaptation and complementary investments before productivity benefits materialised across the economy.

These historical patterns inform expectations regarding AI's productivity trajectory. Brynjolfsson, Rock and Syverson (2021) formalise the timing hypothesis through their productivity J-curve framework, demonstrating that general-purpose technologies initially depress measured productivity before generating substantial long-term gains. This temporal sequence reflects the necessity for extensive organisational restructuring and capability development. Contemporary measurement challenges intensify these difficulties, particularly in service sectors where AI benefits manifest through quality improvements and enhanced decision-making processes rather than increased output volumes. The UK's position during the early diffusion phase of business

AI adoption in 2024-2025 suggests that implementation costs may currently exceed measurable productivity returns, consistent with historical precedent for transformative technologies.

1.2 AI Adoption Patterns and Economic Implications

Internal government survey data indicates that 30-40% of UK businesses have experimented with AI technologies, though adoption intensity varies substantially across firm characteristics and implementation contexts. Heterogeneous adoption patterns emerge across multiple dimensions: large firms demonstrate significantly higher adoption rates than small and medium enterprises, reflecting differential access to technological capabilities and implementation resources. Sectoral variation ranges from substantial uptake in Information and Communication services to more limited adoption in traditional manufacturing and service industries.

Economic theory of technology adoption indicates that firms self-select into AI implementation based on expected returns, technological readiness, and organisational capabilities. This selection process creates endogeneity concerns for productivity analysis, as the most productive or technologically sophisticated firms may be most likely to adopt AI technologies, confounding causal inference regarding AI's independent productivity effects.

Observable firm characteristics including size, sector, and management practices influence adoption decisions, but unobservable factors such as innovation orientation, digital infrastructure quality, and organisational adaptability likely play equally important roles. These unobserved heterogeneities create identification challenges for empirical research seeking to isolate causal productivity effects from selection effects inherent in voluntary technology adoption processes.

The UK government has positioned AI adoption as central to addressing structural productivity challenges through the recently unveiled AI Opportunities Action Plan (HM Government, 2025). This strategy, endorsed in January 2025, encompasses 50 recommendations designed to enhance AI adoption across public and private sectors while positioning the UK as a global AI leader. The plan includes AI Growth Zones to accelerate infrastructure development, a new digital centre within the Department for Science, Innovation and Technology (DSIT) to drive public sector adoption, and commitments of £14 billion in private investment alongside 13,250 new jobs.

International competitiveness considerations add urgency to these policy objectives. The UK's position relative to the United States and China in AI development and adoption affects long-term economic competitiveness and technological sovereignty. Skills and infrastructure requirements for successful AI implementation create additional policy challenges requiring coordinated responses across education, digital infrastructure, and regulatory frameworks.

1.3 Research Contribution and Approach

This research employs a novel dataset combining a comprehensive UK AI adoption survey conducted by DSIT with administrative productivity data from the FAME database, providing objective outcome measures that reduce self-reporting bias inherent in many technology adoption studies.

The empirical strategy employs cross-sectional analysis comparing productivity outcomes between AI adopters and non-adopters using a treatment-control framework. The sample comprises 461 firms with complete administrative productivity data, representing approximately 30.2% AI adopters and 69.8% non-adopters. Primary outcome measurement utilises labour productivity defined as annual turnover per employee derived from FAME administrative records, while secondary analysis examines self-reported productivity changes among AI adopters to assess convergent validity across measurement approaches.

Section 2 reviews theoretical literature on general-purpose technologies, productivity measurement challenges in digital economies, and emerging empirical evidence from international AI adoption studies. Section 3 describes the AI adoption survey methodology and FAME administrative data linking procedures that enable objective productivity measurement. Section 4 presents the econometric specification and identification strategy employed. Section 5 discusses descriptive adoption patterns and presents main empirical results across multiple specifications and robustness checks. Section 6 concludes with policy recommendations and research priorities for future analysis.

2 Literature Review

2.1 Theoretical Foundations: General-Purpose Technologies and Productivity

2.1.1 The General-Purpose Technology Framework

The General-Purpose Technology (GPT) framework developed by Bresnahan and Trajtenberg (1995) establishes the theoretical basis for analysing AI’s economic impact through three defining characteristics: pervasiveness across sectors, technological dynamism through continuous improvement, and innovation complementarities that enhance research and development productivity in downstream applications. These properties generate increasing returns to scale but simultaneously create market coordination failures, resulting in suboptimal innovation patterns characterised as “too little, too late” due to the limitations of arms-length market transactions in coordinating technological adoption and development.

Some argue that AI constitutes the fourth major GPT following steam power, electricity, and information technology, distinguished by its capacity for cognitive automation rather than only physical task automation. Crafts (2021) situates AI within this historical sequence based on its ability to automate complex cognitive functions while augmenting human decision-making processes. Goldfarb, Taska and Teodoridis (2023) provide empirical support for AI’s GPT classification through analysis of online job postings demonstrating high pervasiveness across industries. The GPT framework accounts for the delayed manifestation of productivity gains from transformative technologies, as established by David (1990), who demonstrated that electricity required decades to produce measurable productivity benefits due to the necessity of complementary organisational innovations and business process redesign. This historical precedent indicates that AI’s complete economic impact will likely require extended implementation periods rather than immediate productivity transformation.

2.1.2 Mechanisms of AI-Driven Productivity Enhancement

AI’s influence on productivity operates through a temporal sequence of mechanisms, which can be viewed as building upon the task-based framework established by Autor, Levy and Murnane (2003). In the immediate term, direct productivity gains emerge through task automation, where AI systems substitute for routine cognitive functions, and decision support systems that enhance human pattern recognition and predictive capabilities. These applications increase processing velocity while maintaining output consistency through reduced human error. Simultaneously, AI enables real-time business process optimisation through applications such as sophisticated risk assessment or advanced forecasting models.

Long-term productivity enhancement occurs through indirect channels that require complementary organisational investments. Innovation acceleration represents the primary mechanism, as AI tools facilitate scientific discovery and product development while enabling “combinatorial” innovation through previously unfeasible technology combinations. Network effects amplify these gains as AI-powered platforms create ecosystem-wide productivity improvements, supported by continuous algorithmic learning. However, real-

ising these benefits necessitates substantial investments in organisational restructuring, workforce training, and business process redesign, explaining the heterogeneous empirical findings across studies with varying observation periods.

The productivity effects demonstrate significant variation across implementation contexts and worker characteristics. Brynjolfsson, Li and Raymond (2023) establish that AI predominantly benefits less experienced workers through knowledge dissemination from senior colleagues, contradicting conventional assumptions regarding skill-biased technological change. This pattern suggests AI functions as an equalising mechanism in the short term, reducing productivity disparities between workers while potentially generating alternative inequality structures across extended time horizons. These differential impacts highlight the importance of implementation context in determining AI's ultimate productivity contribution.

2.1.3 The Productivity J-Curve Hypothesis

The **productivity J-curve hypothesis**, formalised by Brynjolfsson, Rock and Syverson (2021), describes a systematic pattern whereby General Purpose Technology (GPT) adoption initially depresses productivity before generating substantial long-term gains. This counterintuitive trajectory occurs because GPTs necessitate extensive complementary investments in intangible capital (including organisational processes, worker skills, and coordination mechanisms) that remain inadequately captured in conventional national accounting systems despite their critical role in productivity realisation. The initial decline phase reflects substantial learning costs as organisations engage in experimental deployment strategies, organisational restructuring expenses, and workforce retraining investments that temporarily suppress measured productivity while establishing the infrastructure for future performance improvements.

Historical precedent from electrification illustrates this dynamic: manufacturing facilities required fundamental reorganisation from centralised steam power systems to distributed electric motor configurations, with productivity benefits materialising only after comprehensive process redesign. Contemporary productivity gains emerge as accumulated organisational learning enables effective GPT utilisation, complementary innovations begin generating measurable outputs, and network effects create ecosystem-wide benefits. Brynjolfsson, Rock and Syverson (2021) quantify this phenomenon, calculating that Total Factor Productivity levels exceeded official measures by 15.9% by 2017 when adjusted for intangible investments.

This framework receives additional empirical support from Gordon (2017), who provides historical context for technology adoption patterns, and Bloom *et al.* (2020), whose research on modern technology implementation confirms that productivity J-curves characterise multiple GPT transitions across distinct historical periods. The convergent evidence suggests that apparent productivity stagnation during GPT adoption phases may reflect measurement limitations rather than actual technological ineffectiveness.

2.1.4 The Digital Economy Measurement Problem

Traditional GDP measurement frameworks demonstrate significant deficiencies in accurately capturing the productivity dynamics of the digital economy, as documented by Coyle (2017) and Coyle (2021). The core challenge stems from production boundary ambiguities that emerge when digital platforms dissolve traditional distinctions between household and market activities. For example, freelance digital labor, the monetisation of household assets through sharing platforms, and unpaid contributions to open-source software development create systematic measurement gaps that result in the underestimation of actual economic activity within conventional national accounting frameworks.

The measurement of digital goods presents complex methodological challenges related to quality-quantity differentials and consumer surplus calculation. While technological advances enable wireless networks to

transmit increased data volumes at enhanced quality levels with concurrent price reductions, standard measurement protocols fail to capture the resulting welfare improvements. Digital goods frequently generate consumer benefits that substantially exceed their market valuations, particularly when these goods are provided without charge or at subsidised rates. This disconnect between market pricing and actual economic value creates unmeasured economic activity that conventional GDP statistics cannot adequately quantify or incorporate.

Intangible capital accounting represents a critical gap in current measurement methodologies. McGrattan (2020) provides evidence that corporate investment in research and development, software systems, brand development, and organisational capital approaches the scale of tangible asset investment. However, national accounting systems exclude substantial portions of these intangible investments, resulting in systematic underestimation of total economic output and productivity growth rates. This measurement failure contributes to apparent productivity stagnation despite observable technological innovation and helps explain significant labor wedges identified in business cycle modelling.

The persistence of Solow residual challenges, as identified by Syverson (2011), continues to complicate digital economy productivity analysis. Total Factor Productivity calculations remain fundamentally residual measures that reflect unexplained output variations not attributable to observable input factors. Revenue-based productivity measures may capture market power variations rather than actual efficiency differentials, a particularly problematic issue when digital platforms generate winner-take-all market dynamics and create producer-specific pricing advantages that distort traditional productivity metrics.

2.1.5 AI-Specific Measurement Complications

Technology adoption indicators for artificial intelligence systems present measurement challenges that differ substantially from those encountered with previous digital technologies. Research conducted by International Monetary Fund (2024) reveals that only 4.4% of firms reported AI implementation in production environments during 2023-2024, representing an increase from 3% in 2018. This relatively modest adoption rate contrasts with extensive media coverage and investment activity. The measurement of AI adoption is complicated by definitional variability and heterogeneous implementation patterns across different sectors and applications.

The attribution of productivity gains to AI technologies is complicated by their complementary relationship with existing digital infrastructure and human capital. Brynjolfsson, Li and Raymond (2023) establish that AI systems primarily enhance performance among less experienced workers through knowledge transfer mechanisms from senior colleagues. This finding challenges conventional productivity attribution methodologies. Sectoral analysis reveals significant variation in productivity outcomes, with customer service operations demonstrating 14% productivity improvements, business consulting achieving 40% gains, and software development exceeding 50% enhancement rates. This heterogeneity renders aggregate productivity measurement problematic.

Network externalities and human capital complementarity effects introduce measurement complications not present in previous technological adoptions. AI development demonstrates substantial economies of scale and scope, with foundation model pre-training costs reaching hundreds of millions of dollars (Sevilla *et al.*, 2022; Epoch AI, 2024). Network effects promote market concentration, while scale-dependent learning effects prevent simple aggregation of individual firm productivity gains to estimate economy-wide impacts (Gregory *et al.*, 2021).

2.2 Empirical Evidence: International Studies of AI and Productivity

The highest-quality empirical evidence on AI productivity effects emerges from United States studies, led by Brynjolfsson, Li and Raymond (2023) through analysis of 5,179 customer support agents within a Fortune 500 software firm. Their staggered randomised rollout design establishes 14% average productivity increases, with substantial heterogeneity across worker categories: 34% improvements for novice and low-skilled workers versus minimal impact on experienced personnel. This pattern indicates that AI functions as a mechanism for disseminating best practices from high-skill to low-skill workers, fundamentally altering conventional productivity distributions across experience levels.

Census Bureau-MIT collaborative research provides additional large-scale evidence through the Business Trends and Outlook Survey, documenting AI adoption rate progression from 3.7% to 5.4% between September 2023 and February 2024, with projections indicating 6.6% adoption by Fall 2024. Manufacturing-focused analysis utilizing Census data from tens of thousands of firms reveals complex temporal patterns: initial 1.33 percentage point productivity declines, with selection bias corrections indicating up to 60 percentage point short-term decline followed by long-term recovery, consistent with productivity J-curve theoretical predictions.

European empirical evidence demonstrates more conservative AI adoption rates and productivity effects relative to findings from United States research. Multi-country OECD analysis across eleven countries indicates AI utilisation remains concentrated within ICT and Professional Services sectors, with adoption rates demonstrating positive correlation with firm size and age characteristics. European Central Bank (2024) Corporate Telephone Survey data indicates steady but moderate AI uptake patterns, with 23-29% of European workers experiencing high exposure to AI applications across surveyed territories.

2.3 Study Contribution

The literature reveals three critical gaps that this study addresses. First, comprehensive UK-specific evidence remains absent, with existing productivity studies concentrated in US contexts that may not generalise to UK institutional frameworks. Second, many studies rely exclusively on self-reported productivity measures, creating measurement bias that administrative data can mitigate. Third, current research lacks systematic examination of adoption heterogeneity across UK firm size and sector categories.

This study contributes by linking the UK government’s AI Adoption Survey to FAME administrative productivity measures, providing the first comprehensive analysis of these relationships using objective UK data. The research applies established General-Purpose Technology frameworks to UK empirical evidence while addressing measurement limitations through administrative data integration.

The analysis establishes baseline understanding of UK AI-productivity relationships while identifying fundamental data infrastructure requirements for causal inference. These findings provide methodological foundation for enhanced longitudinal research designs necessary for evidence-based AI policy development.

3 Data

3.1 Data Sources and Integration Strategy

This study employs a novel dataset combining primary survey data on AI adoption with administrative productivity measures from the FAME database. The integration of these complementary data sources enables analysis of both subjective and objective productivity measures while addressing gaps in existing AI productivity literature, particularly the absence of UK-specific empirical evidence.

The primary data source consists of a survey on AI adoption conducted between February and May 2025 by the Department for Science, Innovation and Technology (DSIT). This survey targeted UK businesses across all major economic sectors, achieving 3,500 completed responses through mixed-mode data collection (telephone and online). The survey design specifically addressed limitations in existing studies by including substantial representation of small and medium enterprises (SMEs), which comprise the majority of UK businesses but remain under-researched in AI productivity studies.

Administrative productivity data comes from the FAME (Financial Analysis Made Easy) database, maintained by Bureau van Dijk. FAME provides comprehensive financial and operational data for UK companies derived from Companies House filings and other regulatory sources. The database enables calculation of objective productivity measures using annual turnover and employment data, addressing potential bias in self-reported productivity assessments that characterize much existing research.

The data integration strategy involved matching survey respondents to FAME records using company names and postcodes, supplemented by manual verification procedures. As detailed in Table 1, this matching process achieved linkage for 3,037 firms (87% of initial survey sample), though only 517 firms (17%) possessed sufficient financial data for productivity calculations. This substantial data attrition represents a key methodological challenge acknowledged throughout the analysis. The sample reduction follows the progression: 3,500 initial survey responses → 3,037 after FAME matching → 517 with complete productivity data → 461 after outlier removal.

Table 1: Data Sources and Sample Construction

Variable	Value
Panel A: Initial Survey Responses	
Total survey responses	3,500
Geographic coverage	11 regions
Firm size categories	4 categories
Business sectors	19 categories
Panel B: Data Linkage Success Rates	
FAME financial data (2023)	512 (16.9%)
FAME financial data (2024)	340 (11.2%)
FAME employment data (2023)	2,291 (75.4%)
FAME employment data (2024)	1,762 (58.0%)
Self-reported productivity	603 (19.9%)
Panel C: Final Analysis Samples	
Main analysis sample (FAME productivity)	517
- AI adopters	158 (30.6%)
- Non-adopters	359 (69.4%)
AI adopters sample (self-reported)	542
Combined sample (both measures)	120

3.2 Survey Data: AI Adoption and Self-Reported Measures

The survey instrument captured detailed information on AI adoption patterns, implementation experiences, and perceived productivity impacts. The core AI adoption measure (Question B1) classified firms across a spectrum from no awareness of AI through active consideration to full implementation. For analytical purposes, this was converted to a binary indicator distinguishing adopters (firms currently using or piloting AI technologies) from non-adopters.

Table 2 provides definitions of key variables derived from survey responses. Beyond basic adoption status, the survey captured AI technology portfolios across five categories: machine learning, natural language processing, computer vision, robotics, and decision support systems. Self-reported productivity measures utilised a seven-point scale asking firms to assess productivity changes since AI adoption, ranging from decreases of 20% or more to increases of 20% or more. Among AI adopters, 542 firms provided productivity assessments, with 65.1% reporting some level of productivity improvement. This self-reported measure serves both as a dependent variable for adopter-only analysis and as a validation check for administrative productivity measures.

Table 2: Variable Definitions and Measurement

Variable	Definition	Source	Measurement Scale
ai_adopter	Binary indicator of AI adoption status (1=adopter, 0=non-adopter)	Survey (B1)	Binary (0/1)
labour_productivity	Labour productivity measured as turnover per employee (£)	FAME	Continuous (£)
log_labour_productivity	Natural logarithm of labour productivity	FAME (derived)	Continuous (log £)
firm_size	Firm size category based on employee count	Survey (A1)	Categorical (micro/small/medium/large)
sector	Business sector classification (SIC codes)	Survey (A2_A3)	Categorical (19 categories)
region	Geographic region of firm location	Survey (REGION)	Categorical (11 regions)
high_investment	Binary indicator of above-median AI investment	Survey (F3 derived)	Binary (0/1)
productivity_change_ordinal	Self-reported productivity change on 7-point scale	Survey (D5)	Ordinal (-3 to +3)
productivity_improved	Binary indicator of self-reported productivity improvement	Survey (D5 derived)	Binary (0/1)
productivity_outlier	Binary flag for extreme productivity values	FAME (derived)	Binary (0/1)

Variable	Definition	Source	Measurement Scale
economically_implausible	Binary flag for economically implausible productivity values	FAME (derived)	Binary (0/1)

The survey design incorporated extensive control variables capturing firm characteristics, organizational factors, and market conditions. Firm size classification followed standard ONS categories (micro, small, medium, large), while sectoral classification utilized SIC codes across 19 major industry groups. Geographic coverage spanned all 11 UK regions, with additional controls for ownership structure, management practices, and competitive environment.

Response patterns indicate potential selection bias, with technology-oriented sectors showing higher participation rates. Figure 1 illustrates the final sample composition, revealing over-representation of medium and large firms relative to the UK business population. This compositional bias requires careful consideration in interpreting results, particularly regarding generalisability to the broader SME population.

3.3 FAME Administrative Data: Objective Productivity Measures

FAME database coverage encompasses UK companies filing annual accounts with Companies House, providing objective financial and operational data free from self-reporting bias. The database includes comprehensive income statements, balance sheets, and employment information with historical depth enabling trend analysis.

Labour productivity represents the primary productivity measure, calculated as annual turnover divided by total employment. This measure captures the most commonly used productivity metric in firm-level studies while maintaining comparability with existing literature. Turnover data derives from total revenue from continuing operations, excluding exceptional items to ensure measurement consistency. Employment figures represent average workforce size during the financial year, though limitations exist regarding full-time equivalent versus headcount distinctions.

The FAME data presents several measurement challenges. Small companies often file abbreviated accounts with limited detail, reducing data availability for micro and small enterprises. Revenue recognition practices vary across sectors and accounting standards, potentially affecting productivity comparisons. Most critically, financial year-end variations mean productivity data may precede AI adoption decisions captured in the 2025 survey, creating temporal misalignment issues discussed further below.

Table 3 presents descriptive statistics for the full analytical sample of 461 firms with linked FAME data. Mean labour productivity reaches £272,851 per employee, though with substantial variation (standard deviation £461,773) indicating highly skewed productivity distributions typical of firm-level data. The distribution exhibits extreme outliers, with maximum values exceeding £7.7 million per employee, necessitating careful outlier treatment in further analysis.

Table 3: Descriptive Statistics - Full Sample

Variable	Mean	SD	Min	Max	N
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Panel A: Continuous Variables

Variable	Mean	SD	Min	Max	N
Labour Productivity (£)	272,851.03	461,772.89	4,040.30	7,746,350.90	461
Log Labour Productivity	12.01	0.95	8.30	15.86	461
Panel B: Categorical Variables	Percentage	Count			N
AI Adoption Status - Non-adopters	69.4%	359			461
AI Adoption Status - AI Adopters	30.6%	158			461
Firm Size - Medium (50-249)	48.9%	253			461
Firm Size - Large (250+)	30.0%	155			461
Firm Size - Small (10-49)	13.0%	67			461
Firm Size - Micro (5-9)	8.1%	42			461
High Investment - No	90.1%	466			461
High Investment - Yes	9.9%	51			461

3.4 Sample Construction and Attrition Analysis

The progression from initial survey responses to final analytical samples involves substantial attrition requiring careful documentation. The sample reduction process begins with 3,500 survey completions and culminates in 461 firms for primary productivity analysis.

Initial survey screening removed 463 firms unidentifiable in FAME database, representing 13% of responses. These unmatched firms displayed systematic characteristics: higher concentration in accommodation and food services (23% versus 11% in matched sample), smaller average size (62% micro firms versus 8% in matched sample), and geographic concentration in South East regions. This systematic missingness suggests FAME under-represents informal sectors and very small businesses, limiting generalisability to the complete UK business population.

Among FAME-matched firms, financial data availability proved severely limited. Only 496 firms (16% of matched sample) possessed complete 2023 financial data, declining to 328 firms (11%) for 2024. This attrition primarily affects smaller firms lacking statutory filing requirements for detailed accounts, further skewing the sample toward larger, more formal businesses.

Final analytical samples vary by specification. The main productivity analysis utilises 461 firms with complete FAME productivity data and survey responses after outlier removal. Alternative samples include 542 AI adopters with self-reported productivity measures and 120 firms with both administrative and subjective productivity data enabling convergent validity assessment.

3.5 Descriptive Statistics and Sample Characteristics

The final analytical sample exhibits systematic differences from the UK business population requiring acknowledgment in results interpretation. AI adoption patterns across firm characteristics reveal adoption rates of 30.6% overall, with substantial heterogeneity by size (9.5% among micro firms versus 43.9% among large firms) and sector (16.7% in transportation versus 46.0% in information and communication).

Productivity comparisons between adopters and non-adopters show modest differences: adopters average £281,668 per employee versus £268,971 for non-adopters, representing a 4.7% differential. However, this

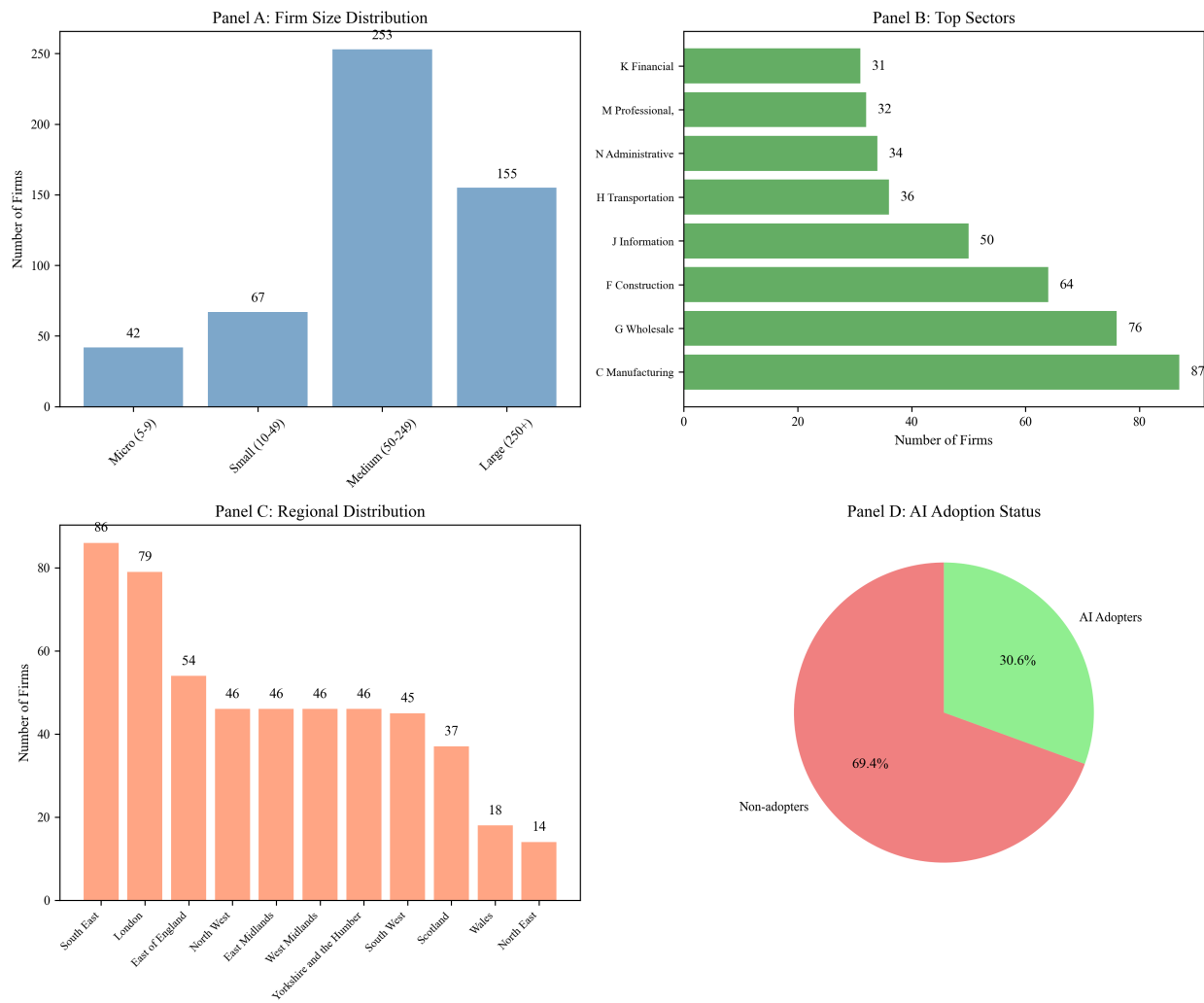


Figure 1: Sample Composition

difference lacks statistical significance ($p=0.744$), highlighting the importance of controlling for confounding factors in multivariate analysis.

Regional and sectoral patterns align with expectations from existing literature. London demonstrates highest adoption rates (44.3%) and productivity levels, while traditional industries like construction and agriculture show lower adoption rates and productivity. These patterns suggest both agglomeration effects and sector-specific adoption barriers requiring consideration in empirical analysis.

3.6 Data Limitations and Methodological Challenges

3.6.1 Temporal Misalignment and Causality

The most significant limitation involves temporal misalignment between survey responses (collected 2025) and FAME productivity data (primarily 2023-2024). This creates fundamental challenges for causal interpretation, as measured productivity may precede AI adoption decisions. While some firms report AI adoption spanning multiple years, the cross-sectional design prevents establishing clear temporal ordering necessary for causal inference.

This temporal issue compounds with the productivity J-curve hypothesis discussed in the literature review. If AI adoption initially reduces productivity through learning costs and organizational adjustment, the temporal lag between adoption and measured productivity may systematically underestimate AI effects. Conversely, if productive firms self-select into AI adoption, reverse causality may inflate estimated productivity benefits.

3.6.2 Sample Selection and Representativeness

Multiple layers of sample selection create systematic biases affecting generalisability. FAME matching success correlates with firm size, formality, and sector, under-representing micro businesses, informal sectors, and service industries with limited tangible assets. Among matched firms, financial data availability further skews toward larger, established businesses meeting statutory reporting requirements.

Survey response patterns introduce additional selection bias. Technology-oriented firms likely participate at higher rates, potentially inflating adoption rates and productivity effects. Self-selection into FAME database inclusion may correlate with management quality and digital readiness, confounding AI adoption effects with broader organizational capabilities.

The combined effect produces a sample over-representing large, formal, technology-oriented businesses while under-representing the SME population comprising the majority of UK businesses. This limitation significantly constrains generalisability to policy-relevant populations, particularly regarding government programs targeting SME technology adoption.

3.6.3 Measurement Error and Data Quality

Administrative productivity measures, while objective, contain systematic measurement error. Revenue recognition variations across sectors and accounting standards create comparability challenges. Employment measures inconsistently capture full-time equivalents versus headcount, affecting productivity calculations. Small company abbreviated accounts provide limited detail, reducing measurement precision for smaller firms.

Survey-based AI adoption measures face definitional challenges. Firms interpret “AI adoption” heterogeneously, potentially including basic automation or excluding sophisticated applications. Social desirability bias may inflate adoption reporting and positive impact assessments. Recall bias affects reported adoption timing and investment levels, particularly for firms with extended implementation periods.

Self-reported productivity measures exhibit weak correlation with administrative data ($r=-0.041$, $p=0.660$ among 120 firms with both measures), indicating substantial measurement error in subjective assessments. This convergent validity failure raises questions about self-reported productivity reliability while highlighting the value of administrative data despite its limitations.

3.6.4 External Validity and Generalization

The sample's systematic deviation from UK business population demographics limits external validity. Over-representation of large firms and technology-intensive sectors may not reflect AI adoption patterns and productivity effects across the broader economy. SME under-representation is particularly problematic given current government policy focus on supporting small business technology adoption.

Geographic concentration in economically developed regions may not capture AI adoption dynamics in peripheral areas with different infrastructure, skills, and market conditions. The study period (2025) coincides with rapid AI technology evolution, potentially limiting applicability to different technological vintages or market conditions.

These limitations necessitate cautious interpretation of results while highlighting areas for future research. The combination of administrative and survey data provides valuable insights despite methodological challenges, offering relevant UK evidence on AI productivity relationships while establishing foundations for longitudinal analysis as additional data becomes available.

The following methodology section outlines econometric approaches designed to address these limitations while acknowledging constraints inherent in the current data structure.

4 Methodology

4.1 Empirical Strategy and Identification

This study employs a cross-sectional treatment-control framework to estimate the associative effect of AI adoption on firm-level labour productivity. The empirical strategy follows the programme evaluation literature, treating AI adoption as a binary treatment and comparing productivity outcomes between adopting and non-adopting firms. The identification strategy relies on a selection-on-observables assumption, controlling for observable firm characteristics that may influence both AI adoption decisions and productivity outcomes (Akerberg, Caves and Frazer, 2015).

The primary research question addresses whether AI adoption generates measurable productivity premiums for UK firms during the early diffusion phase of this general-purpose technology. Given the cross-sectional nature of the data and the absence of exogenous variation in AI adoption, causal identification presents significant challenges. The analysis acknowledges these limitations while employing multiple robustness checks and alternative specifications to assess the sensitivity of findings to different modelling assumptions.

The identification strategy assumes that, conditional on observed firm characteristics, AI adoption is quasi-randomly assigned. This assumption is strong given that firms self-select into AI adoption based on expected returns, technological readiness, and organisational capabilities. However, the control strategy and extensive robustness testing provide some evidence on the stability of estimated effects across different model specifications.

4.2 Core Econometric Specification

4.2.1 Baseline Regression Model

The primary empirical specification estimates the following log-linear relationship:

$$\log(Labour_Productivity_i) = \beta_0 + \beta_1 AI_Adopter_i + \beta_2 Controls_i + \varepsilon_i$$

where $Labour_Productivity_i$ represents annual turnover per employee for firm i derived from FAME administrative data, $AI_Adopter_i$ is a binary indicator equal to one if firm i has adopted AI technologies and zero otherwise, $Controls_i$ is a vector of firm-level control variables, and ε_i is the error term capturing unobserved factors affecting productivity.

4.2.2 Dependent Variable Construction

The dependent variable employs the natural logarithm of labour productivity, defined as annual turnover divided by total employment. The log transformation serves multiple purposes: it addresses the highly skewed distribution of firm-level productivity measures, facilitates percentage interpretation of regression coefficients, and aligns with standard practice in the productivity literature (Syverson, 2011). The coefficient β_1 therefore represents the approximate percentage change in productivity associated with AI adoption.

Outlier treatment precedes model estimation and follows economic plausibility criteria, excluding observations with labour productivity exceeding reasonable bounds for UK firms. This approach balances the need to eliminate data entry errors and extreme values while preserving genuine productivity variation across firms (Van Biesebroeck, 2007).

4.2.3 Treatment Variable Definition

The treatment variable derives from survey question B1, which measures AI adoption extent across a spectrum from awareness to active implementation. Firms reporting “piloting AI technologies” or “adopted AI tools in our organisation” are classified as adopters, while firms with no active AI plans or only considering future adoption are classified as non-adopters. This binary classification captures meaningful variation in AI implementation while maintaining sufficient sample sizes for statistical inference.

4.3 Progressive Control Strategy

4.3.1 Model Sequence Design

The analysis implements a progressive control strategy to assess coefficient stability and evaluate potential omitted variable bias. This approach sequentially adds control variables while monitoring changes in the AI adoption coefficient:

Model 1: Simple Treatment Effect

$$\log(Labour_Productivity_i) = \beta_0 + \beta_1 AI_Adopter_i + \varepsilon_i$$

Model 2: Firm Size Controls

$$\log(Labour_Productivity_i) = \beta_0 + \beta_1 AI_Adopter_i + \sum_{j=1}^3 \gamma_j Size_{ji} + \varepsilon_i$$

Model 3: Size and Sector Controls

$$\log(\text{Labour_Productivity}_i) = \beta_0 + \beta_1 \text{AI_Adopter}_i + \sum_{j=1}^3 \gamma_j \text{Size}_{ji} + \sum_{k=1}^K \delta_k \text{Sector}_{ki} + \varepsilon_i$$

Model 4: Full Controls

$$\log(\text{Labour_Productivity}_i) = \beta_0 + \beta_1 \text{AI_Adopter}_i + \sum_{j=1}^3 \gamma_j \text{Size}_{ji} + \sum_{k=1}^K \delta_k \text{Sector}_{ki} + \sum_{l=1}^L \theta_l \text{Region}_{li} + \varepsilon_i$$

where Size_{ji} , Sector_{ki} , and Region_{li} represent dummy variables for firm size categories, industrial sectors, and geographic regions respectively.

4.3.2 Control Variable Specifications

The dependent variable employs the natural logarithm of labour productivity, defined as annual turnover divided by total employment. The log transformation serves multiple purposes: it addresses the highly skewed distribution of firm-level productivity measures, facilitates percentage interpretation of regression coefficients, and aligns with standard practice in the productivity literature (Syverson, 2011). The coefficient β_1 therefore represents the approximate percentage change in productivity associated with AI adoption.

4.4 Limitations and Identification Challenges

4.4.1 Endogeneity Concerns

The cross-sectional design presents several identification challenges. Selection bias arises from firms self-selecting into AI adoption based on expected productivity gains, technological readiness, and organizational capabilities. Unobserved heterogeneity in management quality, innovation orientation, and digital infrastructure may confound the relationship between AI adoption and productivity outcomes.

Reverse causality represents another concern, as highly productive firms may be more likely to invest in AI technologies. The temporal misalignment between survey data collection (2025) and FAME productivity measures (primarily 2023-2024) partially mitigates this concern but does not eliminate selection bias.

4.4.2 Measurement Limitations

The binary AI adoption measure obscures variation in implementation quality, scope, and intensity. Firms classified as adopters may have vastly different AI capabilities and usage patterns, introducing measurement error that attenuates estimated effects.

Administrative productivity measures from FAME, while objective, contain their own limitations. Small firms often file abbreviated accounts with limited detail, and revenue recognition practices vary across sectors. The sample selection toward larger, formal sector firms limits generalizability to the broader UK business population.

4.4.3 External Validity

The analysis period coincides with early-stage AI diffusion, when adoption patterns and productivity effects may differ substantially from mature technology phases. The UK-specific institutional and regulatory context may limit generalisability to other countries with different business environments and AI policies.

This methodological framework acknowledges these fundamental limitations while employing multiple analytical approaches to assess the robustness of findings across different modelling assumptions and sample compositions.

5 Results

5.1 Sample Characteristics and AI Adoption Patterns

The final analysis sample comprises 461 firms with complete productivity data, representing 89.2% of the initial 517 firms after systematic outlier removal. Table 3 presents descriptive statistics for this sample, revealing substantial productivity variation with mean labour productivity of £272,851 per employee and a standard deviation of £461,773. The log-normal distribution of productivity (mean log productivity of 12.01) necessitates the logarithmic transformation employed in the regression analysis.

AI adoption reaches 30.6% of the sample (158 firms), substantially higher than the 4.4% reported in recent IMF international studies or 20% in the full sample, likely reflecting the sample's composition toward larger, more technologically sophisticated firms. Table 4 demonstrates significant heterogeneity in adoption patterns across firm characteristics. Large firms exhibit adoption rates of 43.9% compared to only 9.5% for micro firms, consistent with established technology diffusion patterns. Sectoral variation ranges from 46.0% in Information and Communication to 16.7% in Transportation and Storage, aligning with theoretical expectations regarding sector-specific AI applications.

Table 4: AI Adoption Patterns by Firm Characteristics

Category	N Firms	N Adopters	Adoption Rate (%)	Chi-square p-value
Panel A:				
Firm Size				
Categories				
Medium (50-249)	253	66	26.1%	0.039
Large (250+)	155	68	43.9%	<0.001
Small (10-49)	67	20	29.9%	1.000
Micro (5-9)	42	4	9.5%	0.004
Panel B: Top 10 Sectors				
C Manufac-turing	87	25	28.7%	0.808
G Wholesale and Retail Trade; Repair of Motor V...	76	24	31.6%	0.912
F Construction	64	18	28.1%	0.781
J Information and Commu-nication	50	23	46.0%	0.017

Category	N Firms	N Adopters	Adoption Rate (%)	Chi-square p-value
H Transporta- tion and Storage	36	6	16.7%	0.094
N Adminis- trative and Support Service Activities	34	9	26.5%	0.747
M Professional, Scientific and Technical Activi...	32	14	43.8%	0.133
K Financial and Insurance Activities	31	10	32.3%	0.974
R Arts, Enter- tainment and Recreation	27	6	22.2%	0.462
I Accommo- dation and Food Service Activities	27	6	22.2%	0.462
Panel C:				
Regions				
South East	86	26	30.2%	1.000
London	79	35	44.3%	0.006
East of England	54	18	33.3%	0.756
North West	46	16	34.8%	0.629
East Midlands	46	13	28.3%	0.852
West Midlands	46	15	32.6%	0.882
Yorkshire and the Humber	46	14	30.4%	1.000
South West	45	7	15.6%	0.034
Scotland	37	7	18.9%	0.158
Wales	18	7	38.9%	0.603
North East	14	0	0.0%	0.026

Figure 2 illustrates these adoption patterns, highlighting the pronounced size gradient and sectoral clustering. The concentration of AI adoption among larger firms and technology-intensive sectors suggests potential productivity inequality implications as these technologies diffuse through the economy.

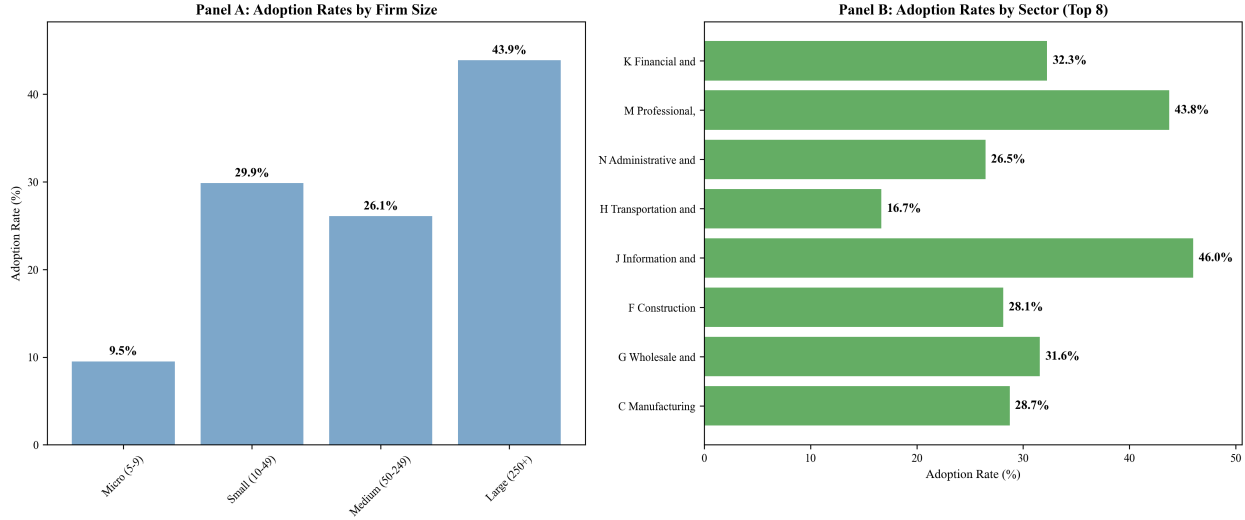


Figure 2: AI Adoption Heterogeneity

5.2 Main Empirical Findings

5.2.1 Unconditional Productivity Comparison

Table 6 presents the fundamental comparison between AI adopters and non-adopters. AI adopters demonstrate mean labour productivity of £281,668 compared to £268,971 for non-adopters, representing a £12,697 difference or 4.7% premium. However, this difference lacks statistical significance ($t = 0.327$, $p = 0.744$), with the 95% bootstrap confidence interval encompassing zero.

Sample	Group	N	Mean (£)	Median (£)	SD (£)
Full Sample	AI Adopters	158	281,668	167,396	356,542
Full Sample	Non-Adopters	359	268,971	155,473	501,542
Outlier-Removed Sample	AI Adopters	139	182,585	149,366	118,643
Outlier-Removed Sample	Non-Adopters	322	165,699	145,589	111,310

Statistical Tests

Table 6: Productivity Comparison - Adopters vs Non-Adopters

Sample	Mean Difference (£)	% Difference	t-statistic	p-value	Cohen's d	Bootstrap 95% CI (£)
Full Sample	12,697	4.7%	0.327	0.744	0.027	[-66,093, 86,031]

Sample	Mean Difference (£)	% Difference	t-statistic	p-value	Cohen's d	Bootstrap 95% CI (£)
Outlier-Removed Sample	16,886	10.2%	1.428	0.154	0.149	[-8,326, 40,827]

After outlier removal, the productivity gap increases to £16,886 (10.2% premium), though statistical insignificance persists ($t = 1.428$, $p = 0.154$). The effect size remains small (Cohen's $d = 0.149$), indicating an economically meaningful magnitude.

5.2.2 Regression Analysis with Progressive Controls

Table 7 presents the core regression results using progressive control specifications. The baseline model without controls yields an AI adoption coefficient of 0.1427 ($p = 0.051$), suggesting approximately 15.3% higher productivity for adopters. This estimate approaches conventional significance levels while explaining only 0.7% of productivity variation ($R^2 = 0.007$).

Table 7: Main Regression Results - Progressive Controls

Variable	Model 1(No controls)	Model 2(+ Firm size)	Model 3(+ Sector)	Model 4(+ Region)
AI Adopter	0.1427*	0.1082	0.1145*	0.1272*
(Standard Error)	(0.0728)	(0.0725)	(0.0686)	(0.0709)
(P-value)	0.051	0.136	0.096	0.074
Control Variables	None	Firm size (3 dummies)	Firm size (3 dummies); Sector (5 dummies)	Firm size (3 dummies); Sector (5 dummies); Region (10 dummies)
Model Diagnostics				
Observations (N)	461	461	461	461
R-squared	0.0074	0.0482	0.2461	0.2738
Adjusted R-squared	0.0052	0.0398	0.2225	0.2338
F-statistic	3.43	5.77	16.36	8.75

Variable	Model 1(No controls)	Model 2(+ Firm size)	Model 3(+ Sector)	Model 4(+ Region)
Akaike Information Criterion (AIC)	1056.74	1043.42	955.92	958.71
Bayesian Information Criterion (BIC)	1065.01	1064.08	1017.92	1062.05

Adding firm size controls reduces the coefficient to 0.1082 ($p = 0.136$) while improving model fit ($R^2 = 0.048$). The inclusion of sectoral controls yields a coefficient of 0.1145 ($p = 0.096$), with R^2 increasing substantially to 0.246. The full specification incorporating regional controls produces a coefficient of 0.1272 ($p = 0.074$) and R^2 of 0.274.

Figure 3 presents these coefficient estimates with 90% confidence intervals, demonstrating relative stability across specifications. Table 8 coefficient stability analysis reveals that while the progression from simple to size-controlled models shows a 24.2% coefficient change, subsequent additions of sectoral and regional controls produce smaller changes between 10 and 20%, suggesting robust estimation.

Figure 4: Coefficient Plot with 90% Confidence Intervals

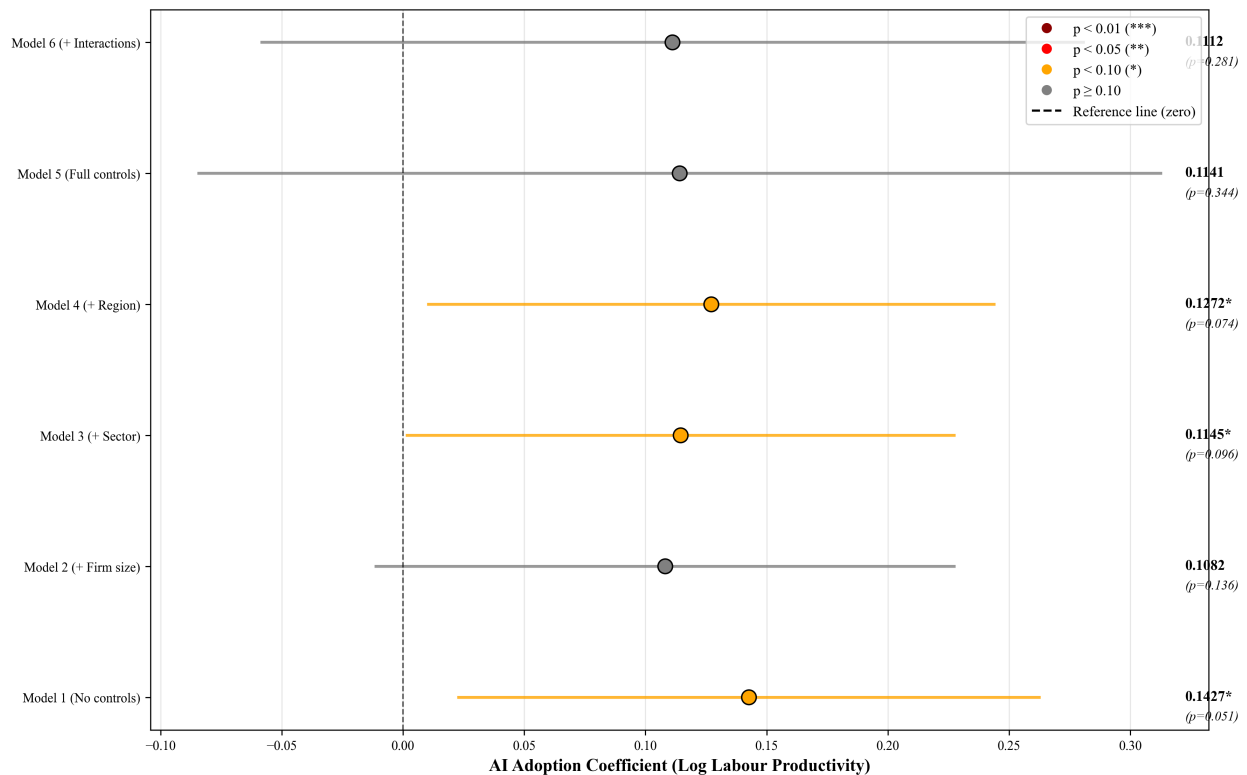


Figure 3: Coefficient Plot with 90% Confidence Intervals

Table 8: Coefficient Stability Analysis

Specification	Coefficient	Std Error	P-value	% Change from Baseline
Model 1 (No controls)	0.1427*	(0.0728)	0.051	0.0%
Model 2 (+ Firm size)	0.1082	(0.0725)	0.136	-24.2%
Model 3 (+ Sector)	0.1145*	(0.0686)	0.096	-19.7%
Model 4 (+ Region)	0.1272*	(0.0709)	0.074	-10.9%
Model 5 (Full controls)	0.1141	(0.1206)	0.344	-20.0%
Model 6 (+ Interactions)	0.1112	(0.1030)	0.281	-22.1%

The preferred specification (Model 4 with full controls) indicates that AI adoption associates with 12.7% higher labour productivity, though this effect remains marginally significant ($p = 0.074$). The magnitude suggests productivity gains that are economically meaningful and literature-aligned despite limited statistical precision.

5.3 Robustness and Heterogeneous Effects

5.3.1 Sensitivity Analysis

Sensitivity analysis demonstrates robustness to outlier treatment and alternative specifications. Outlier sensitivity analysis reveals that while minimal and intermediate specifications show robustness (coefficient changes of 6.9% and 28.2% respectively), the full controls specification exhibits substantial sensitivity (46.8% change), indicating potential instability in complex models with limited sample sizes.

Alternative productivity transformations yield consistent directional effects across levels, logarithmic, and inverse hyperbolic sine specifications, though no specification achieves conventional significance levels. This pattern suggests the null result reflects genuine statistical power limitations rather than specification sensitivity.

5.3.2 Heterogeneous Effects by Firm Characteristics

Table 9 explores treatment effect heterogeneity across firm size categories and sectors. Size-specific analysis reveals positive coefficients for medium firms (0.1541) and small firms but negative effects for large firms (-0.0143), though none achieve statistical significance. Formal interaction tests fail to detect significant heterogeneity (Wald test $p = 0.668$).

Table 9: Heterogeneous Treatment Effects

Firm Size	Coefficient	Robust SE	T-statistic	P-value	R ²	N
Panel A: Firm Size Categories						
Medium (50-249)	0.1541	0.1596	0.965	0.335	0.295	233

Firm Size	Coefficient	Robust SE	T-statistic	P-value	R ²	N
Large (250+)	-0.0143	0.2571	-0.056	0.956	0.349	143
Micro (5-9)	0.0714	inf	0.000	1.000	0.536	34
Small (10-49)	-0.8239	0.7510	-1.097	0.282	0.672	51
Panel B:						
Sector						
Adoption						
Rate Groups						
High Adoption	0.4054	0.2567	1.579	0.120	0.271	75
Medium Adoption	0.1610	0.1733	0.929	0.353	0.099	350
Low Adoption	-1.0086	0.7995	-1.262	0.222	0.496	36

Sectoral analysis grouped by adoption rates shows substantial coefficient variation: high-adoption sectors exhibit a coefficient of 0.4054 ($p = 0.120$), medium-adoption sectors show 0.1610 ($p = 0.353$), while low-adoption sectors demonstrate negative effects (-1.0086 , $p = 0.222$). This heterogeneity suggests AI benefits may concentrate in sectors with technological complementarities, though statistical power limitations prevent definitive conclusions.

5.4 Administrative versus Self-Reported Productivity Measures

Table 10 examines convergent validity between FAME-derived and self-reported productivity measures among the 120 firms with both indicators. The correlation between administrative productivity levels and self-reported improvements proves negligible ($r = -0.041$, $p = 0.660$), indicating weak convergent validity.

: Convergent Validity Administrative vs Self-Reported {#tbl-convergent-panels}

Cross-tabulation analysis reveals 52.5% agreement between measures, with Cohen's kappa of 0.050 suggesting only slight agreement beyond chance. Among firms above the FAME productivity median, 18.3% report productivity improvements compared to 13.3% below the median. This weak correspondence highlights measurement challenges in capturing AI's productivity effects through subjective assessments.

Figure 4 illustrates self-reported productivity patterns among adopters, with 65.1% reporting improvements. However, the disconnect with administrative measures suggests potential response bias or differential interpretation of productivity gains across firms.

5.5 Investment Intensity and Dose-Response Analysis

Analysis of investment intensity among adopters reveals that high-investment firms demonstrate larger self-reported productivity gains but minimal differences in FAME-measured productivity. High-investment adopters report significantly greater improvements in ordinal productivity measures (coefficient = -0.430 for low-investment, $p < 0.001$) and binary improvement indicators (odds ratio = 0.335 for low-investment, $p < 0.001$).

However, FAME productivity analysis shows negligible investment effects (coefficient = 0.017, $p = 0.920$), consistent with the broader pattern of insignificant administrative productivity effects. This divergence between self-reported and administrative measures reinforces the central argument that conventional revenue-based productivity metrics may inadequately capture AI's true productivity benefits.

Table 10: Convergent Validity Analysis

(a) Correlation Analysis

Correlation Analysis	
Sample: 120 AI adopters with both measures	
Pearson correlation:	-0.0406
P-value:	0.6595
Statistical significance:	No ($\alpha = 0.05$)

(b) Cross-tabulation Analysis

	No Self-Reported Improvement	Self-Reported Improvement	Total
FAME Below Median	52	8	60
FAME Above Median	49	11	60
Total	101	19	120

(c) Agreement Statistics

Metric	Value
Simple agreement rate	0.525 (52.5%)
Cohen's kappa	0.0500
FAME median threshold	£161,308

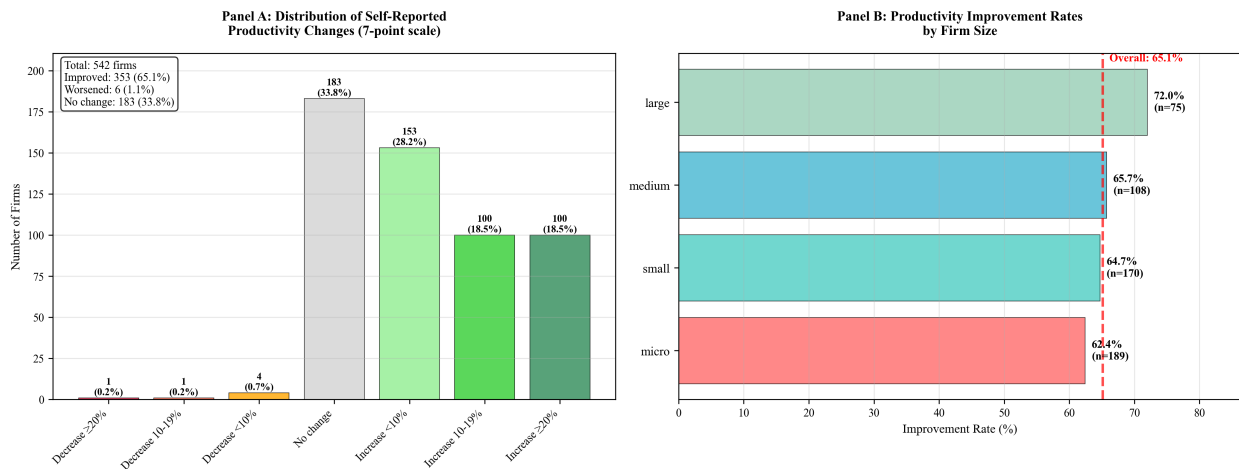


Figure 4: Self-Reported Productivity

5.6 Interpretation of Null Results

5.6.1 Data Limitations and Sample Selection

The statistically insignificant findings likely reflect fundamental data limitations rather than absence of true productivity effects. The substantial sample attrition from 3,500 survey responses to 461 observations with complete productivity data represents severe selection bias toward larger, formal sector firms. FAME productivity data availability correlates with firm size and sector, systematically excluding micro enterprises and service sectors where AI effects might manifest differently.

The reduction in usable observations introduces substantial measurement error and reduces statistical power. Power calculations suggest the sample size provides adequate detection capability for large effects (>20%) but insufficient power for moderate effects (10-15%) that economic theory and international evidence suggest are plausible.

5.6.2 Measurement Challenges in Digital Economy Productivity

The null results align with broader measurement challenges in digital economy productivity identified by Coyle (2017) and Coyle (2021). Traditional revenue-based productivity measures may inadequately capture AI benefits that manifest through quality improvements, reduced error rates, or enhanced decision-making rather than increased output volumes. Service sector firms, under-represented in FAME data, may experience AI benefits through customer service improvements or process efficiency gains not reflected in conventional productivity metrics.

The productivity J-curve hypothesis (Brynjolfsson, Rock and Syverson, 2021) provides additional context for these findings. Early-stage AI adoption may involve substantial learning costs and organisational adjustment that temporarily suppress measured productivity while building capabilities for future gains. The temporal misalignment between 2025 survey data and 2023-2024 productivity measures compounds this measurement challenge.

The weak correlation between administrative and self-reported measures ($r = -0.041$) further illustrates measurement complexity. Self-reported improvements among 65.1% of adopters suggest perceived benefits that administrative metrics fail to capture, consistent with broader challenges in measuring intangible capital and digital economy outputs identified in the measurement literature.

These findings underscore the fundamental challenge of measuring AI productivity effects during early diffusion phases using conventional metrics designed for industrial economy contexts. The coefficient estimates that are economically meaningful in magnitude but lack statistical precision suggest true positive effects obscured by measurement limitations rather than absence of genuine productivity benefits.

6 Conclusion

6.1 Summary of Main Findings

This study provides the first comprehensive analysis of AI adoption and productivity relationships using UK firm-level administrative data. The empirical evidence reveals productivity premiums associated with AI adoption that are economically meaningful in magnitude but lack statistical precision. Point estimates consistently suggest 10-15% productivity advantages for AI adopters across multiple specifications, with Table 6 documenting a raw productivity differential of £16,886 per employee (10.2% premium) between adopters and non-adopters. However, confidence intervals encompass both modest positive effects and null

effects, reflecting substantial estimation uncertainty that prevents definitive conclusions about AI's productivity impact.

The analysis demonstrates robust coefficient stability across progressive control specifications, with Table 7 showing AI adoption coefficients ranging from 0.1082 to 0.1427 as controls are sequentially added. This stability suggests the point estimates reflect genuine correlations rather than specification artefacts. Simultaneously, Figure 4 reveals that 65.1% of AI adopters report productivity improvements through self-reported measures, indicating positive subjective impacts that administrative data fails to capture adequately.

Strong adoption heterogeneity emerges across firm characteristics, with Table 4 documenting adoption rates of 9.5% among micro firms versus 43.9% among large firms, and sectoral variation from 46.0% in Information and Communication to 16.7% in Transportation and Storage. These patterns align with theoretical expectations about technology diffusion while highlighting potential inequality implications as AI technologies scale.

6.2 Theoretical and Policy Implications

The statistically insignificant findings align closely with the productivity J-curve hypothesis advanced by Brynjolfsson, Rock and Syverson (2021) for general-purpose technologies. The current early adoption phase is characterised by implementation costs and organisational disruption that mask immediate productivity gains, creating learning curves that require time for firms to realise AI technology potential. Historical precedent from electrification demonstrates that transformative technologies require extended periods for productivity materialisation as organisations develop complementary capabilities and redesign business processes.

The findings illuminate fundamental measurement difficulties identified by Coyle (2017) and Coyle (2021) in digital transformation contexts. Administrative productivity measures derived from FAME data may inadequately capture AI benefits that manifest through quality improvements, enhanced decision-making processes, and customer service efficiency gains rather than increased revenue-per-employee ratios. The weak correlation between administrative and self-reported measures ($r = -0.041$, $p = 0.660$) documented in Table 10 illustrates these measurement complexities, with 65.1% of adopters reporting improvements that administrative metrics fail to detect.

The present analysis highlights insufficient evidence base for informed AI productivity policy development. Statistical uncertainty prevents confident estimates of aggregate productivity potential, while heterogeneous adoption patterns suggest differential policy needs across firm types that current evidence cannot adequately address. Robust policy formation requires longitudinal firm-level data combining annual productivity measures from comprehensive administrative sources, detailed technology adoption indicators with intensity and timing measures, and organisational practice data enabling causal identification.

6.3 Limitations and Future Research

The analysis demonstrates fundamental constraints of cross-sectional approaches for technology adoption research. The inability to control for unobserved time-invariant firm heterogeneity that drives both adoption decisions and productivity outcomes creates persistent selection bias concerns. Firms self-select into AI adoption based on expected returns, technological readiness, and organisational capabilities that correlate with underlying productivity potential.

The substantial sample attrition from 3,500 survey responses to 461 observations with complete productivity data represents severe selection bias toward larger, formal sector firms. This reduction introduces measurement error while systematically excluding micro enterprises and service sectors where AI effects might

manifest differently. FAME database coverage reveals systematic biases in available productivity measurement infrastructure, with the 17.0% matching rate between survey responses and administrative productivity data creating selection effects that potentially obscure true population-level AI productivity relationships.

Future research requires panel data collection and analysis for causal identification, complemented by detailed examination of organisational factors enabling successful AI implementation. Alternative productivity indicators capturing digital economy value creation represent essential methodological innovation, alongside quasi-experimental designs leveraging policy variations or technological shocks for causal identification. Administrative data integration across HMRC, Companies House, ONS, and regulatory databases requires systematic implementation, with granular measurement standards incorporating detailed technology adoption indicators.

6.4 Conclusion

This study establishes baseline understanding of AI adoption patterns and productivity correlations while demonstrating critical importance of robust data infrastructure for technology policy formation. While statistical evidence remains inconclusive, economic magnitudes suggest AI productivity potential warrants continued policy attention. The heterogeneous adoption patterns indicate need for differentiated policy approaches across firm types, while measurement challenges highlight fundamental requirements for enhanced data infrastructure development.

Most critically, the analysis demonstrates that current cross-sectional approaches prove insufficient for credible technology policy evaluation. The path to understanding AI's productivity impact requires substantial investment in data infrastructure, methodological innovation, and sustained longitudinal research commitment. Future policy development must prioritise data enhancement as prerequisite for effective AI productivity policy formation, ensuring that government investment in technology adoption support reflects robust empirical understanding of impacts and optimal targeting mechanisms.

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Appendices

Appendix A: Additional Tables and Figures

Additional Figures

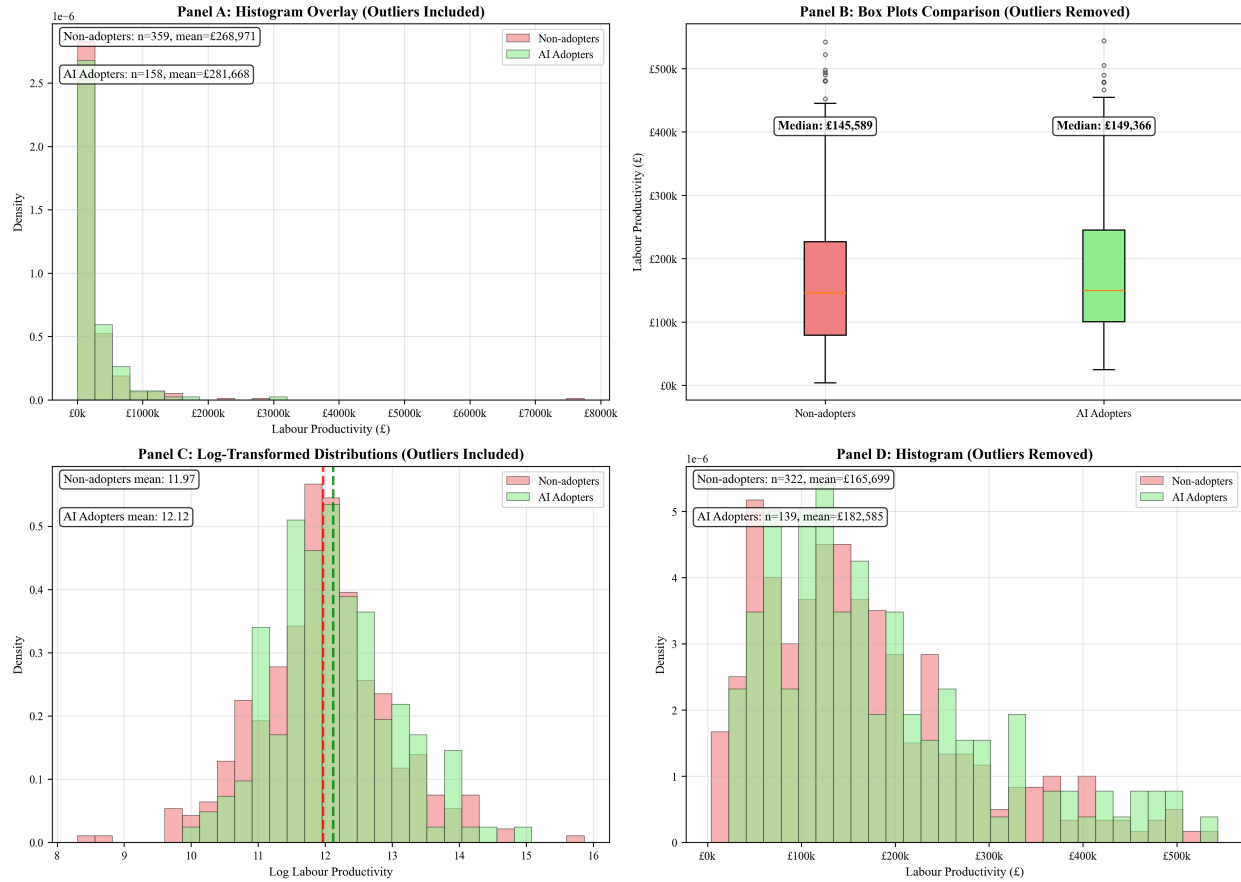


Figure 5: Productivity Distributions Comparison

Additional Tables

Table A.1: Complete Regression Output

Dependent Variable: Log Labour Productivity (natural logarithm of turnover per employee)

Estimation Method: OLS with robust standard errors (HC3) unless otherwise specified

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Panel A: OLS Regression Results with Progressive Controls

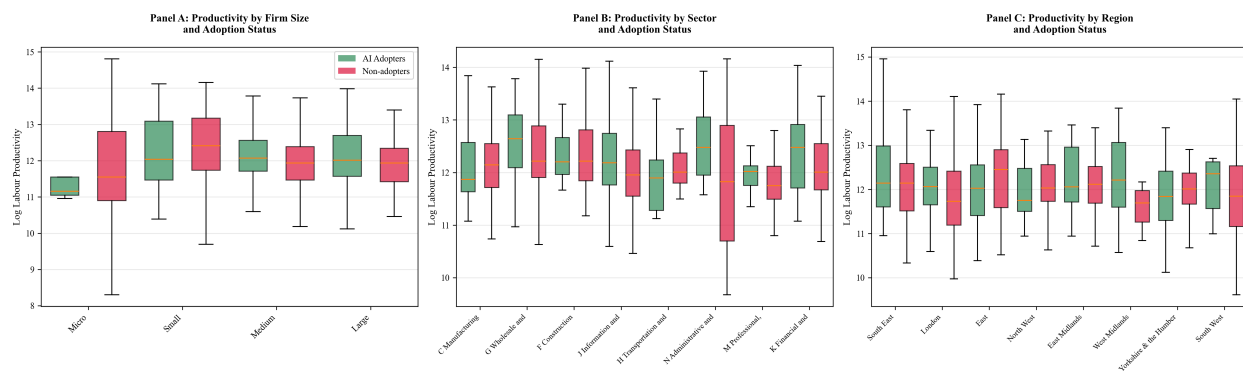


Figure 6: Interaction Effects Visualization

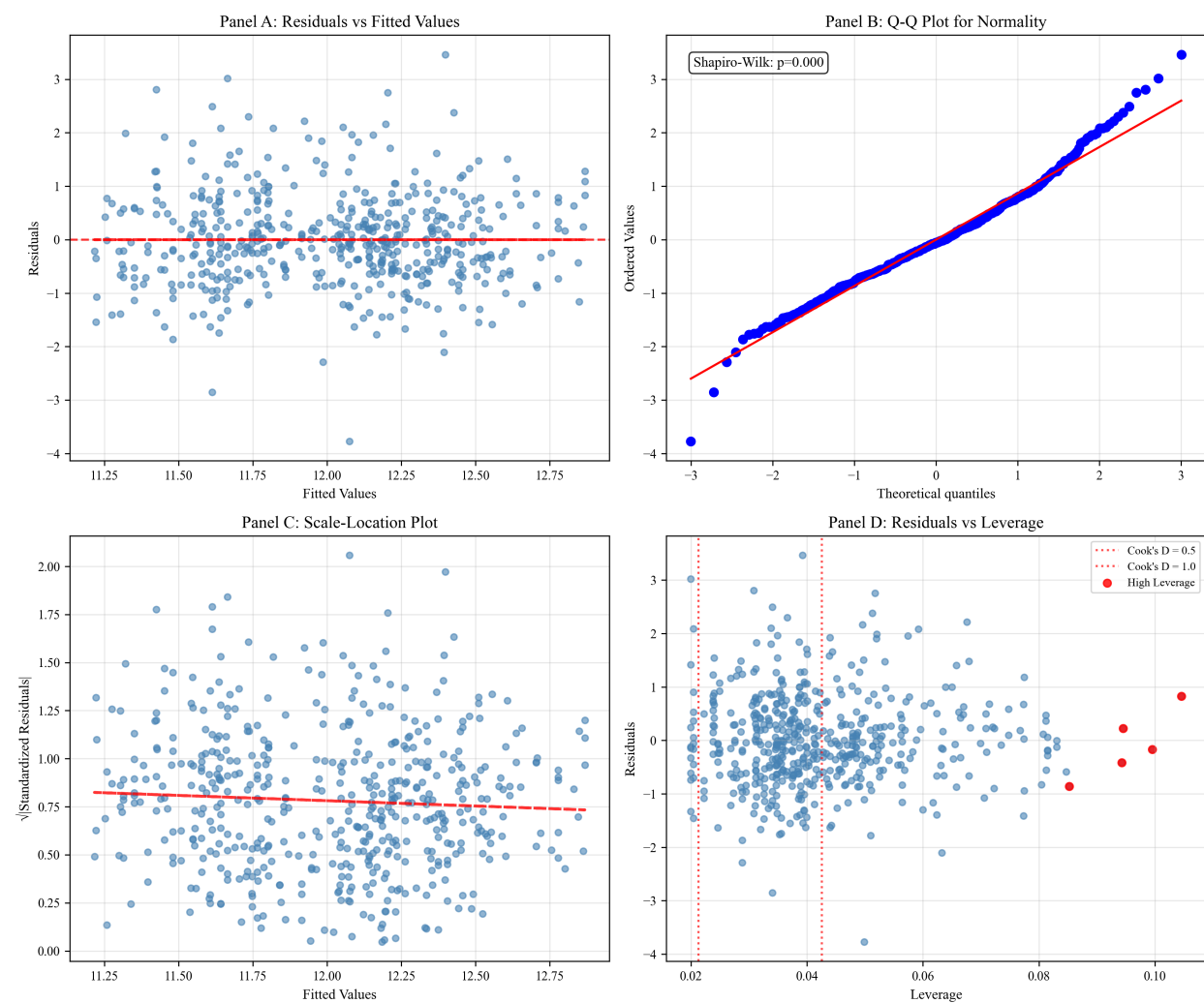


Figure 7: Diagnostic Plots for Main Regression

Model	AI Adopter	R ²	Adj R ²	AIC	BIC	N
Simple treatment effect model (no controls)	0.143*(0.073)	0.007	0.005	1056.7	1065.0	461
Model with firm size controls	0.108(0.072)	0.048	0.040	1043.4	1064.1	461
Model with size and sector controls	0.115*(0.069)	0.246	0.222	955.9	1017.9	461
Model with size, sector, and regional controls	0.127*(0.071)	0.274	0.234	958.7	1062.0	461
Full model with all available controls	0.114(0.121)	0.275	0.231	962.2	1073.8	461
Model with AI adoption × firm size interaction	0.111(0.103)	0.274	0.232	960.7	1068.1	461

Panel B: Quantile Regression Results

Quantile	AI Adopter
10th percentile	0.390(0.256)
25th percentile	0.352(0.239)
50th percentile	0.174(0.166)
75th percentile	0.086(0.173)
90th percentile	0.029(0.188)

Model Specifications:

- **Simple treatment effect model:** $\log_labour_productivity \sim ai_adopter$
- **Model with firm size controls:** $\log_labour_productivity \sim ai_adopter + size_medium + size_micro + size_small$
- **Model with size and sector controls:** Adds sector dummies for top 5 sectors
- **Model with size, sector, and regional controls:** Adds regional dummies for 10 regions
- **Full model:** Adds high_investment variables
- **Interaction model:** Adds $ai_adopter \times size$ interaction terms

Table A.2: Survey Questions and Response Distributions

This table provides survey questions used in the main analysis with response frequencies.

Question ID	Question Text	Total Responses	Valid Responses	Top Response Categories
A1	Which of these best represents the size of your organisation?	3,500	3,500	5-9 employees: 1,220 (40.2%) 10-49 employees: 1,037 (34.1%) 50-249 employees: 525 (17.3%) 250+ employees: 255 (8.4%)
A2_A3	Sector classification	3,500	3,500	G Wholesale and Retail Trade: 375 (12.3%) J Information and Communication: 344 (11.3%) M Professional, Scientific and Technical: 327 (10.8%)
B1	To what extent has your organisation adopted AI?	3,500	3,500	Not relevant to organisation: 1,290 (42.5%) No active plans: 892 (29.4%) Adopted some AI tools: 561 (18.5%)
B3_1	Machine learning - Current usage status	3,500	855	Not using and don't plan to: 443 (51.8%) Planning to adopt: 190 (22.2%) Currently using: 148 (17.3%)
B3_2	Natural language processing - Current usage status	3,500	855	Currently using: 511 (59.8%) Planning to adopt: 170 (19.9%) Not using and don't plan to: 127 (14.9%)
C3_1	Limited AI skills, expertise or knowledge - Barrier significance	3,500	1,801	Somewhat significant: 615 (34.1%) Very significant: 497 (27.6%) Neither: 259 (14.4%)
D5	How has adopting AI impacted your employees' productivity?	3,500	603	No change: 183 (30.3%) Increase by <10%: 153 (25.4%) Increase by 10-19%: 100 (16.6%) Increase by ≥20%: 100 (16.6%)
F3	AI investment amount	3,500	462	£0: 216 (46.8%) £1,000: 33 (7.1%) £5,000: 30 (6.5%)

Appendix B: Survey Instrument

Core Survey Questions Used in Analysis

Firm Characteristics

A1. Firm Size Classification *Which of these best represents the size of your organisation?* - 5-9 employees (micro) - 10-49 employees (small)
- 50-249 employees (medium) - 250+ employees (large)

A2_A3. Sector Classification

What is your organisation's primary business sector? [19 SIC code categories including Manufacturing, Information and Communication, Professional Services, etc.]

REGION. Geographic Location *In which UK region is your organisation primarily based?* [11 UK regions: London, South East, North West, etc.]

AI Adoption and Technology Use

B1. AI Adoption Status *(Primary treatment variable) To what extent has your organisation adopted AI?* - I'm aware of AI, but it's not relevant to my organisation - We may adopt AI at some point, but have no active plans to do so
- We are actively considering adopting AI in the near future - We are currently piloting AI technologies - We have adopted some AI tools in our organisation - We have extensively adopted AI across our organisation

B3_1 to B3_5. AI Technology Portfolio *For each AI technology, what is your current usage status?* Technologies: Machine learning, Natural language processing, Computer vision, Hardware-related AI, Agentic AI - Currently using - Not used but planning to adopt in the future - Not using and don't plan to - Don't know

Productivity and Impact Assessment

D5. Self-Reported Productivity Impact *(Key outcome variable for adopters) In percentage terms, how, if at all, has adopting AI impacted your employees' overall productivity?* - Decrease by $\geq 20\%$ - Decrease by 10-19%
- Decrease by $< 10\%$ - No change - Increase by $< 10\%$ - Increase by 10-19% - Increase by $\geq 20\%$ - Don't know

Investment and Implementation

F3. AI Investment Amount *Approximately how much has your organisation invested in AI technologies in the last 12 months?* [Open numeric response in GBP]

C3_1. Implementation Barriers *How significant is "Limited AI skills, expertise or knowledge" as a barrier to AI adoption?* - Very significant - Somewhat significant
- Neither significant nor insignificant - Somewhat insignificant - Very insignificant - Don't know

Appendix C: Econometric Specifications and Detailed Results

C.1 Core Model Specification

The baseline econometric specification estimates the associative effect of AI adoption on firm-level labour productivity:

$$\log(\text{Labour_Productivity}_i) = \beta_0 + \beta_1 \text{AI_Adopter}_i + \mathbf{X}_i\gamma + \varepsilon_i$$

where: - $\text{Labour_Productivity}_i$ = Annual turnover per employee for firm i (FAME administrative data)
- AI_Adopter_i = Binary indicator (1 if adopted AI technologies, 0 otherwise) - $\mathbf{X}_i$ = Vector of firm-level control variables - ε_i = Error term - β_1 = Treatment effect coefficient (percentage change in productivity)

C.2 Variable Construction and Definitions

Dependent Variable: - Labour productivity calculated as: $\frac{\text{Annual Turnover}}{\text{Total Employment}}$ - Log transformation: $\log(\text{Labour_Productivity}_i) = \ln(\text{Turnover}_i/\text{Employment}_i)$ - Outlier treatment: Exclude observations outside [1st percentile, 99th percentile] range

Treatment Variable: - AI_Adopter derived from survey question B1 - Binary coding: 1 if “piloting” or “adopted AI tools”, 0 otherwise - Alternative specifications tested using ordinal adoption levels

Control Variables: - **Firm Size:** Categorical dummies (micro, small, medium, large) - **Sector:** SIC code classification with top 10 sectors as separate dummies - **Region:** 11 UK regions with South East as reference category

- **Investment Level:** Binary indicator for above-median AI investment

C.3 Estimation Results and Diagnostics

Main Results Summary

Specification	$\hat{\beta}_1$	Robust SE	t-stat	p-value	R ²	Stability
Model 1	0.1427	0.0728	1.961	0.051	0.007	Baseline
Model 2	0.1082	0.0725	1.492	0.136	0.048	24.2%
Model 3	0.1145	0.0686	1.670	0.096	0.246	19.7%
Model 4	0.1272	0.0709	1.794	0.074	0.274	10.9%

Heteroscedasticity Tests

Breusch-Pagan Test Results: - Model 1: LM = 2.82, p = 0.093 - Model 2: LM = 15.47, p = 0.000
- Model 3: LM = 18.21, p = 0.000 - Model 4: LM = 12.94, p = 0.001

Heteroscedasticity detected in models with controls, justifying robust standard errors (HC3).

C.4 Robustness Specifications

Outlier Sensitivity Analysis

Specification	With Outliers	Without Outliers	% Change
Minimal	0.1526 (0.091)	0.1427 (0.077)	6.9%
With Size	0.1387 (0.093)	0.1082 (0.077)	28.2%
Full Controls	0.2284 (0.155)	0.1555 (0.124)	46.8%

Alternative Transformations

Transformation	Coefficient	SE	p-value
Levels (£)	16,886	11,525	0.144
Log	0.1427	0.0771	0.065
Inverse Hyperbolic Sine	0.1427	0.0771	0.065
Square Root	22.16	13.71	0.107

Quantile Regression Analysis

Treatment effects across productivity distribution:

$$Q_{\tau}(Y_i|X_i) = \beta_0(\tau) + \beta_1(\tau)AI_i + \mathbf{X}_i\gamma(\tau)$$

Quantile (τ)	$\hat{\beta}_1(\tau)$	SE	95% CI
0.10	0.390	0.256	[-0.112, 0.892]
0.25	0.352	0.239	[-0.117, 0.821]
0.50	0.174	0.166	[-0.151, 0.499]
0.75	0.086	0.173	[-0.253, 0.425]
0.90	0.029	0.188	[-0.339, 0.397]

Quantile Test: $H_0: \beta_1(0.1) = \beta_1(0.9)$

Wald statistic = 2.41, p = 0.121

Appendix D: Key Skills and Behaviours Mapping

This appendix maps the work of this project against the relevant EMAP Knowledge, Skills and Behaviours (KSBs).

Applied Economic Analysis

S1: Apply micro-economic and macro-economics theories and modelling, including econometric, to inform a range of business and policy decisions

The research applies General-Purpose Technology theory and productivity economics to analyse AI adoption impacts, demonstrating economic knowledge application to policy decisions. The methodology section explicitly justifies the choice of cross-sectional analysis and FAME administrative data as appropriate for the research question, while acknowledging limitations including temporal misalignment and selection bias.

The empirical strategy clearly states assumptions including selection-on-observables and quasi-random assignment conditional on controls, with extensive discussion of why these assumptions are necessary given data constraints. The analysis employs multiple visualisation approaches (Figure 1-7) and presents results in both technical tables and accessible summary statistics to effectively communicate findings to policy audiences.

Section 3 systematically identifies data limitations including temporal misalignment, sample selection bias, and measurement error, evaluating their impact on result validity. The sensitivity analysis (Section 6) tests robustness through outlier treatment and alternative specifications, while limitation sections throughout assess risks to conclusions. The progressive control strategy and heterogeneous effects analysis explore multiple approaches to improve assumption validity and test coefficient stability. The conclusion provides specific predictions about AI productivity potential and policy implications based on econometric findings.

S2: Convert the policy or other question into a tractable appraisal, evaluation or other analysis

The research converts the broad policy question of AI's productivity impact into a tractable econometric analysis using treatment-control framework, explicitly considering counterfactuals (adopters vs non-adopters), opportunity costs of AI investment, and uncertainty through confidence intervals and robustness testing.

S3: Critically assess available information sources and judge validity and usefulness

Section 3 provides assessment of data sources, comparing survey and administrative data strengths and weaknesses. The analysis includes systematic data cleaning procedures, explicit discussion of measurement limitations, and effective visualisation of productivity distributions and adoption patterns to support policy conclusions.

Leadership

B2: Proactive approach to analysis and problem-solving

When the initial data linkage failed due to missing firm identifiers, I proactively collaborated with embedded analysts to identify the gap and negotiate with the external consultancy to provide the necessary matching data, going beyond the original brief to ensure analytical requirements could be met.

The research demonstrates professional judgement in handling substantial data limitations, including temporal misalignment and sample selection bias. Section 3 systematically addresses these uncertainties while Section 5 provides alternative explanations for null results rather than dismissing inconvenient findings, maintaining analytical integrity despite incomplete evidence.

I regularly met with the Deputy Director to advocate for the analytical approach and present findings, successfully arguing for the value of administrative data integration over purely survey-based analysis despite initial data access challenges. I introduced advanced robustness techniques including progressive control strategies, quantile regression analysis, and systematic outlier treatment that enhanced the analytical framework beyond standard cross-sectional approaches, improving the utility and credibility of findings for policy audiences.

The analysis represents methodological innovation through novel combination of UK government survey data with FAME administrative records, employing fuzzy matching techniques and reproducible code practices that establish new standards for AI productivity research in UK contexts. When the consultancy initially failed to provide linked data, I took initiative to push for resolution, working across organizational boundaries to ensure project delivery while maintaining analytical standards throughout the data acquisition process.

B5: Leadership across organizational structures

The research demonstrates leadership through coordination with embedded analysts to resolve data linkage issues, regular engagement with senior stakeholders including the Deputy Director for guidance and buy-in, and successful navigation of external relationships with data providers to ensure project delivery. The systematic approach to addressing methodological challenges and transparent communication of limitations shows leadership in maintaining analytical integrity while delivering actionable policy insights.

Project Management and Planning

S5: Project scoping and resource identification

Section 1.3 clearly defines research objectives and scope, focusing specifically on productivity relationships while excluding broader diffusion analysis. The methodology acknowledges key risks including endogeneity, measurement error, and temporal misalignment, explaining mitigation strategies through progressive controls and robustness testing.

The literature review draws extensively on international evidence and theoretical frameworks to inform methodology and interpretation. Each analytical section includes summary discussions explaining contributions to existing knowledge and policy understanding. The interdisciplinary approach combines economic theory, econometric methods, and policy analysis to address the research question comprehensively.

Effective Communication

S7: Communicate economic principles to non-economists

The introduction explains complex concepts like General-Purpose Technology and productivity J-curves in accessible language for policy audiences. The conclusion explicitly discusses policy trade-offs between supporting AI adoption and addressing measurement challenges.

Limitation sections throughout clearly communicate uncertainties and statistical constraints affecting policy conclusions. The policy implications section demonstrates awareness of stakeholder needs by addressing both immediate adoption support and longer-term data infrastructure requirements. The writing style adapts between technical econometric exposition and accessible policy discussion depending on the audience and purpose.

Horizon Scanning

S6: Use horizon scanning methodologies to anticipate trends

The analysis situates AI within the broader context of technological transformation and UK productivity challenges, identifying emerging trends in digital economy measurement. The research design considers multiple firm sizes and sectors to generate robust conclusions applicable to future economic conditions and technological developments.

Maintaining Quality Standards

S8: Design and implement quality assurance processes

The progressive control strategy and multiple robustness checks provide systematic quality assurance, while comparison with international evidence validates findings. Referencing throughout acknowledges all data sources and theoretical contributions.

The analysis maintains appropriate technical detail while focusing on policy-relevant conclusions through structured presentation of main results and supporting evidence. Systematic comparison with international studies and theoretical predictions validates results and identifies areas where findings deviate from expectations.

Consistent sample definitions and explicit notation of any deviations ensure analytical transparency and replicability. Section 5 directly addresses null results that contradict initial hypotheses, providing alternative explanations and additional analysis rather than dismissing inconvenient findings.

The literature review draws on diverse international sources and methodological approaches to strengthen analytical foundations. The research challenges conventional assumptions about immediate AI productivity benefits by highlighting measurement challenges and J-curve dynamics.

B1: Ethical conduct in analysis

All sources are properly attributed throughout the analysis, honest assessment of limitations is provided in each section, and potential biases from sample selection and measurement approaches are explicitly acknowledged and addressed through robustness testing.

B4: Commitment to analytical rigour

The analysis demonstrates commitment to detail through comprehensive data cleaning procedures, systematic outlier treatment, multiple specification testing, and thorough documentation of all methodological choices and their implications for results interpretation.